Oxford College of Emory University

Chapter 14, Part 2

Money Creation and Bank Balance Sheets

Guilherme Paiva Pinto

October 29, 2026

Roadmap

1. Loan creation and payment settlement
2. Repayment, withdrawals, and losses
3. A conditional deposit-expansion model
4. Practice and an individual checkpoint

Bank Accounting

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,000
Loans: 900	Equity: 100
Total: 1,100	Total: 1,100

Reserves and loans are assets. Deposits are liabilities. Equity is the residual.

Loan Origination

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,000
Loans: 900	Equity: 100
Total: 1,100	Total: 1,100

The bank credits a new $120 loan to a customer's deposit.

Loan Origination

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,000
Loans: 900	Equity: 100
Total: 1,100	Total: 1,100

The bank credits a new $120 loan to a customer's deposit.

Loans: 900 + 120 = 1,020. Deposits: 1,000 + 120 = 1,120.

Loan Origination

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,000
Loans: 900	Equity: 100
Total: 1,100	Total: 1,100

The bank credits a new $120 loan to a customer's deposit.

Loans: 900 + 120 = 1,020. Deposits: 1,000 + 120 = 1,120.

Reserves remain 200; equity remains 100.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

Payment to Another Bank

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,120
Loans: 1,020	Equity: 100
Total: 1,220	Total: 1,220

The borrower pays $120 to a seller at Bank B.

Payment to Another Bank

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,120
Loans: 1,020	Equity: 100
Total: 1,220	Total: 1,220

The borrower pays $120 to a seller at Bank B.

Bank A: reserves −120; deposits −120.

Payment to Another Bank

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,120
Loans: 1,020	Equity: 100
Total: 1,220	Total: 1,220

The borrower pays $120 to a seller at Bank B.

Bank A: reserves −120; deposits −120.

Bank B: reserves +120; seller’s deposits +120.

The Banking System

Change since before the loan	Bank A	Bank B	Combined
Reserves	−120	+120	0
Loans	+120	0	+120
Deposits	0	+120	+120

The payment moves deposits; the loan created the additional $120 of deposits.

Cash Withdrawals

- A customer withdraws $50: bank reserves −50 and deposits −50.

Cash Withdrawals

- A customer withdraws $50: bank reserves −50 and deposits −50.
- Public currency rises by $50, offsetting the fall in deposits.

Cash Withdrawals

- A customer withdraws $50: bank reserves −50 and deposits −50.
- Public currency rises by $50, offsetting the fall in deposits.
- Currency + deposits is unchanged; the bank has less liquidity.

Principal Repayment

- A borrower uses $50 in a deposit at the same bank to repay principal.

Principal Repayment

- A borrower uses $50 in a deposit at the same bank to repay principal.
- Loans fall by $50 and deposits fall by $50.

Principal Repayment

- A borrower uses $50 in a deposit at the same bank to repay principal.
- Loans fall by $50 and deposits fall by $50.
- Currency + deposits falls by $50; reserves and equity do not change.

Loan Losses and Bank Capital

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,000
Loans: 900	Equity: 100
Total: 1,100	Total: 1,100

The bank writes off $80 of unrecoverable loans.

Loan Losses and Bank Capital

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,000
Loans: 900	Equity: 100
Total: 1,100	Total: 1,100

The bank writes off $80 of unrecoverable loans.

Loans fall to 820; equity falls to 20.

Loan Losses and Bank Capital

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,000
Loans: 900	Equity: 100
Total: 1,100	Total: 1,100

The bank writes off $80 of unrecoverable loans.

Loans fall to 820; equity falls to 20.

Deposits and reserves remain unchanged at the write-off.

Constraints on Lending

- **Capital**: losses and additional assets affect the equity cushion.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

Constraints on Lending

- **Capital**: losses and additional assets affect the equity cushion.
- **Liquidity and funding**: payments can leave the bank.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

Constraints on Lending

- **Capital**: losses and additional assets affect the equity cushion.
- **Liquidity and funding**: payments can leave the bank.
- **Risk and profitability**: borrowers must qualify, and lending must cover costs.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

A Reserve-Ratio Model

- Let banks hold reserves equal to **10% of deposits**: $r = 0.10$.

A Reserve-Ratio Model

- Let banks hold reserves equal to **10% of deposits**: $r = 0.10$.
- All loan proceeds are redeposited; no extra reserves or currency leakage.

A Reserve-Ratio Model

- Let banks hold reserves equal to **10% of deposits**: $r = 0.10$.
- All loan proceeds are redeposited; no extra reserves or currency leakage.
- Banks lend each available amount; borrowers want the loans; other constraints do not bind.

Successive Deposit Rounds

Round	New deposit ($)	Reserves retained	New loan
1	100.00	10.00	90.00
2	90.00	9.00	81.00
3	81.00	8.10	72.90

The central bank first buys a $100 security from a nonbank seller, adding reserves and a deposit.

The Deposit Multiplier

$$\Delta D = 100 + 90 + 81 + \cdots = \frac{100}{0.10} = 1{,}000$$

- **Total new deposits**: \$1,000, including the initial \$100.

The Deposit Multiplier

$$\Delta D = 100 + 90 + 81 + \cdots = \frac{100}{0.10} = 1{,}000$$

- **Total new deposits**: \$1,000, including the initial \$100.
- **Additional loans**: \$900. Total extra reserves: \$100.

The Deposit Multiplier

$$\Delta D = 100 + 90 + 81 + \cdots = \frac{100}{0.10} = 1{,}000$$

- **Total new deposits**: \$1,000, including the initial \$100.
- **Additional loans**: \$900. Total extra reserves: \$100.
- These are limiting model results under the stated conditions.

Depositing Existing Currency

A customer instead deposits $100 of currency already held outside banks.

Depositing Existing Currency

A customer instead deposits $100 of currency already held outside banks.

Initial change: currency −100; deposits +100. No initial increase in currency + deposits.

Depositing Existing Currency

A customer instead deposits $100 of currency already held outside banks.

Initial change: currency −100; deposits +100. No initial increase in currency + deposits.

With the same expansion conditions, deposits rise by 1,000 in total, but currency + deposits rises by 900.

The Model and U.S. Banking

- The U.S. reserve requirement has been **0% since March 26, 2020**.

Source: Federal Reserve Board, Reserve Requirements.

The Model and U.S. Banking

- The U.S. reserve requirement has been **0% since March 26, 2020**.
- Banks still need liquidity and capital and must manage funding and credit risk.

Source: Federal Reserve Board, Reserve Requirements.

The Model and U.S. Banking

- The U.S. reserve requirement has been **0% since March 26, 2020**.
- Banks still need liquidity and capital and must manage funding and credit risk.
- $1/r$ at $r = 0$ does not predict infinite lending or an infinite money supply.

Source: Federal Reserve Board, Reserve Requirements.

Central-Bank Security Purchases

Transaction	Reserves	Nonbank deposits
Buy from a bank	Bank reserves rise	No immediate required change
Buy from a nonbank seller	Seller's bank reserves rise	Seller's deposit rises

Both transactions exchange assets; the immediate effect depends on the seller.

Balance-Sheet Practice

Use the worksheet's bank: reserves 150, loans 750, securities 100, deposits 900, equity 100.

Record a new $90 loan, then its payment to another bank.

For each event, verify assets = liabilities + equity and state what happens to total deposits.

Bank Accounting Checkpoint

A bank writes off $30 of loans, with no repayment or deposit withdrawal. What changes immediately?

Bank Accounting Checkpoint

A bank writes off $30 of loans, with no repayment or deposit withdrawal. What changes immediately?

Loans −30 and equity −30.

Bank Accounting Checkpoint

A bank writes off $30 of loans, with no repayment or deposit withdrawal. What changes immediately?

Loans −30 and equity −30.

Deposits and reserves do not change solely because of the write-off.

Banking and Monetary Policy

- Bank lending changes deposits; settlement moves reserves.

Banking and Monetary Policy

- Bank lending changes deposits; settlement moves reserves.
- The multiplier depends on its model conditions.

Banking and Monetary Policy

- Bank lending changes deposits; settlement moves reserves.
- The multiplier depends on its model conditions.
- Chapter 15 connects central-bank policy, interest rates, spending, and AD.